

Prisha Priyadarshini

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EDUCATION

Rutgers University – New Brunswick

Bachelor of Science in Computer Science, Minor in Mathematics GPA: 3.50/4.00

New Brunswick, NJ

Expected May 2027

RESEARCH EXPERIENCE

Lead ML Researcher / Rutgers MBS Exchange

Jan. 2026 – Present

The Center for Discrete Mathematics & Theoretical Computer Science (DIMACS)

New Brunswick, NJ

- Led an empirical study under Dr. Linda Ness comparing interpretable decision tree models (**SPLIT**, **LicketySPLIT**, **LicketyRESPLIT**, **GOSDT**) with SOTA boosting methods (**XGBoost**, **CatBoost**, **LightGBM**) across 6 datasets.
- Demonstrated that preprocessing with **ThresholdGuessBinarizer (TGB)** eliminates performance gaps between boosting models and interpretable decision trees, achieving comparable accuracy while preserving interpretability.
- Analyzed the impact of **SMOTE** on **Rashomon set size**, showing that class balancing significantly expands model equivalence classes.
- Quantified **performance–complexity tradeoffs** using accuracy, recall, runtime, and Rashomon set size, finding that increased model complexity yields marginal performance gains but substantially larger Rashomon sets.

AI Researcher (Multi-Agent Deliberation & Consensus Dynamics)

Sept. 2025 – Present

Algoverse AI

Remote

- Built a multi-agent framework to study consensus formation, showing that **model-to-model deference drives convergence over independent reasoning**.
- Ran large-scale 20-round deliberation experiments across **GPT-4.1**, **Mistral**, **LLaMA**, and **Phi** models on subjective and objective benchmarks.
- Introduced quantitative metrics (Inter-agent disagreement rate, model deference rate) to analyze interaction dynamics, identifying **hierarchical influence patterns (up to 80% small to large model deference)**.
- Developed a rotation-based framework to separate model identity from response quality, revealing that **large-to-small model deference rates increase relative to baseline**, slightly weakening hierarchical influence patterns.
- Showed that prompting strategies can control multi-agent behavior, **increasing disagreement and reducing deference, with implications for system reliability**.
- In submission to 2 ICML workshops (Pluralistic Alignment and AI4GOOD).

AI Researcher (Multimodal Machine Learning)

June 2025 – Dec. 2025

Algoverse AI

Remote

- Contributed to a hierarchical scene-captioning pipeline **integrating dynamic stride window selection and multimodal chain-of-thought (MMCoT) reasoning** for temporally coherent caption generation.
- Designed and implemented evaluation frameworks (BLEU-4, METEOR, CIDEr, BERTScore) and conducted ablation studies on frame sampling (**5, 20, 40**) and model aggregators (**Phi**, **Mistral**, **Qwen3**).
- Ran large-scale YouCook2 experiments on 30 RTX A6000 GPUs and achieved **17% higher CIDEr than GPT-4o** and **14% over VLLaMA-3**.
- Accepted for an **oral presentation at NeurIPS 2025 (7HVU)** and **accepted to AAAI 2026 (AI4EDU)**.

Algorithms Research Shadow

Jan. 2025 – May 2025

The College of New Jersey

Ewing, NJ

- Researched classical and modern folding algorithms under Dr. Dimitris Papamichail (Nussinov, Zuker, LinearFold, MCTS).
- Implemented sparse dynamic programming to prune redundant structures and reduce runtime.
- Conducted SLURM-based large-scale experiments using ViennaRNA across thousands of RNA sequences.

PUBLICATIONS

• Directional Influence and Consensus Formation in Multi-Agent Systems

Prisha Priyadarshini, Aryan Shrivastava

In submission to ICML 2026 AI4GOOD and Pluralistic Alignment

• DynaStride: Dynamic Stride Windowing with MMCoT for Instructional Multi-Scene Captioning

Eddison Pham, Prisha Priyadarshini, Adrian Maliackel, Kanishk Bandi, Cristian Meo, Kevin Zhu

NeurIPS 2025 7HVU (Oral), AAAI 2026 AI4EDU

INDUSTRY EXPERIENCE

Incoming Machine Learning Intern

Regeneron

May 2026

Troy, NY

- Incoming Summer 2026

AI/ML Open Source Contributor (DeepChem)

Google Summer of Code

May 2026

Remote

- Selected as **1 of 5 out of 200+ applicants** for DeepChem to integrate **OLMo 7B models** into its LLM infrastructure
- Extending the **HuggingFace** wrapper with **generation, batching, and causal LM support**, enabling scalable training and inference for large decoder-only models such as OLMo.

AI Associate Developer

Insurity – SpatialKey Team

Oct. 2025 – Jan. 2026

Remote

- Built **geospatial preprocessing pipelines over 6M+ records**, integrating climate and peril data via time alignment and spatial filtering.
- Engineered features using Pandas and GeoPandas to support ML models.
- Built and evaluated LightGBM-based multi-class peril classifiers; **applied SMOTE class balancing, increasing accuracy and F1 score by 10%**.
- Experimented with a Temporal Fusion Transformer (TFT) model, modeling seasonal dependencies via cyclical week encoding and trained GPU-accelerated PyTorch models in a managed Anaconda environment on NVIDIA RTX hardware.
- Additionally, developed a geospatial change-detection system for before/after wildfire satellite imagery using OpenCV and U-Net++ for high-resolution segmentation.

PROJECTS

Knowledge Distillation for Reasoning in MoE and Dense LLMs | *Python, PyTorch, HuggingFace* April. 2026

- Developed an offline knowledge distillation pipeline using **LoRA (< 1% trainable parameters)** to transfer chain-of-thought reasoning and token-level distributions from large LLMs to a 3B student via **sequence and logit distillation (KL divergence + temperature scaling)**, achieving **4.5% accuracy gains** and **96%+ BERTScore on SciBench and TheoremQA**.

GenreBlender | *Python, PyTorch, Streamlit* Feb. 2026

- Engineered a Generative AI system combining Meta's MusicGen and a **4-layer PyTorch Multilayer Perceptron (92% validation accuracy)** to create and quantitatively evaluate controllable music genre blends via a weighted probability framework in a Streamlit app.

PocketRAG | *Python, Flask, FAISS, Gemini, AWS, Docker, RAG* Sept. 2025

- Engineered a production RAG pipeline with document chunking, **FAISS vector retrieval**, and Gemini inference; containerized and deployed on **AWS EC2 (Docker + Gunicorn)** achieving **sub-second latency**.

PROFESSIONAL DEVELOPMENT & EXTRACURRICULARS

Workshop Reviewer

May 2026

ICML 2026 Pluralistic Alignment Workshop

- Reviewed and evaluated research submissions for the ICML 2026 Pluralistic Alignment Workshop through the OpenReview peer review process.

Mathematics Grader

Dec. 2025 – Present

Rutgers University – New Brunswick

- Grade homework and exams for Multivariable Calculus across 3 sections.

Peer Tutor

Sept. 2024 – May 2025

The College of New Jersey

- Provided one-on-one tutoring for Data Structures, Algorithms, Linear Algebra, and Statistics.

Recruitment Chair

Dec. 2024 – May 2025

Kappa Theta Pi

- Led recruitment efforts and onboarded 10 new members for Spring 2025.

Student Athlete (NCAA Division III Tennis)

Aug. 2023 – May 2024

The College of New Jersey

- Balanced 20+ hrs/week training with academics; earned All-NJAC recognition.

TECHNICAL SKILLS

AI & ML: PyTorch, Hugging Face, OpenCV, Transformers, scikit-learn

Research Domains: Multi-agent Systems, Interpretable ML, Multimodal ML

Programming Languages: Python, C, C++, JavaScript, Java

Libraries: NumPy, Pandas, GeoPandas, Matplotlib, Seaborn

Cloud, Systems, & MLOps: GitHub, AWS, Azure, Docker, CI/CD, pytest, Jira, Anaconda, Vercel

Certifications: AWS Certified Machine Learning — Specialty, AWS Certified Cloud Practitioner